

DOGMA-Gen: Short-Horizon Dog Motion Generation from Wearable IMUs

Jiatong Guo (gjt@bu.edu), Maolin Wei (mlwei@bu.edu)

Abstract

Dog-worn inertial sensors make it possible to observe motion outside the laboratory, but generating future dog motion from sparse, noisy IMU streams remains difficult without synchronized motion-capture supervision. We study an IMU-only future-motion generation problem: given 6s of raw dog IMU history, predict a plausible 2s future clip in a derived pseudo-pose space. We introduce DOGMA-Gen, a reproducible staged pipeline that converts 20Hz IMU windows into stern-centered pseudo-pose with root heading and translation, trains temporal VAEs in both pseudo-pose and raw-IMU spaces, and evaluates conditional latent diffusion and DiT-style generators under a shared prediction and plausibility protocol. This design keeps raw IMU as the sensor-space condition while using pseudo-pose as the common motion representation for reconstruction, generation, and analysis. On held-out windows, pseudo-pose latents preserve the motion representation far better than raw-IMU latents projected back to pseudo-pose space, reducing MPJPE from 129.92 mm to 7.55 mm and heading error from 39.16° to 1.69°. Across generation ablations, pose-space targets yield stronger paired prediction accuracy than raw-IMU target branches, while no single model dominates all realism, smoothness, and diversity metrics. These results provide a reproducible baseline and empirical design evidence for scaling wearable-sensor dog motion synthesis toward motion-prior data for quadruped robotic systems. Project page: <https://github.com/guojiatong/DOGMA>

1. Introduction

Generating dog motion from wearable inertial sensors offers a scalable alternative to controlled motion-capture collection. Dogs move through diverse indoor and outdoor environments, interact with people and objects, and exhibit behavior that is difficult to reproduce in a laboratory setting. The DOGMA dataset provides a rare source of such data: approximately 20 hours of synchronized egocentric video and full-body IMU recordings from 19 dog-human pairs in natural environments. Our goal is to convert these sensor streams into plausible short-horizon motion clips that can

support dog motion synthesis and, eventually, motion-prior learning for quadruped robots.

As shown in Fig. 1, the sensing setup uses 10 dog-mounted IMUs distributed over the head, trunk, and limb segments. Rather than attempting full dog motion capture, we focus on a narrower and more measurable task: generating a plausible 2s future motion sequence from a 6s history of raw IMU signals. The input is a 20Hz raw IMU tensor, and the output is a pseudo-pose sequence containing 10 stern-centered relative joint positions, root heading, and 3D root translation. This formulation avoids claiming motion-capture-grade dog pose while testing whether real dog IMU histories contain enough short-term dynamics to condition future motion generation.

The task is technically challenging because sparse IMU signals are locally informative but globally under-constrained. Human sparse-IMU work has shown that a small set of body-worn inertial sensors requires temporal priors, kinematic constraints, and calibration care to recover plausible full-body motion [5, 17, 20, 21]. The dog setting adds additional difficulty: sensor attachment is less rigid, dog morphology differs from human skeletons, and reliable dog motion-capture supervision is rare. Consequently, directly treating raw IMU as a pose target can produce signals that are numerically plausible in sensor space but difficult to inspect as biological motion.

The key design choice in DOGMA-Gen is to condition on raw IMU while generating and evaluating in a pseudo-pose motion space. The pipeline preprocesses DOGMA IMU streams into 20Hz windows, derives pseudo-pose sequences through an IMU2Pose visualization transform, and trains generative models to predict future motion from raw IMU history. A temporal VAE compresses 8 s pseudo-pose windows into compact latent sequences, and conditional diffusion models sample future motion latents conditioned on the past IMU window. We also train raw-IMU VAE and diffusion ablations to test whether sensor-space generation is a competitive alternative.

This staged design separates three questions. First, can a pseudo-pose VAE reconstruct held-out dog motion well enough to serve as a latent target? Second, does generating raw IMU remain useful after projecting the result back to pseudo-pose? Third, do diffusion models improve distributional realism without sacrificing temporal or kinematic



Figure 1. **10-IMU Dog Wearable Sensor Setup.** Each dog is instrumented with 10 IMUs covering the head, trunk, and limb segments. We use these synchronized sensors to construct a 130-dimensional raw IMU feature per frame.

plausibility? We therefore evaluate reconstruction accuracy and generative quality using MPJPE, heading error, ADE, FDE, Motion-FID, Diversity@10, and dog-specific plausibility checks.

The experiments show that representation choice has a large effect on performance. On all test VAE windows, the pseudo-pose VAE achieves 7.55 mm MPJPE and 1.69° heading error, while a raw-IMU VAE projected back to pseudo-pose space gives 129.92 mm and 39.16° . In the six-model diffusion comparison, pose-space output branches are substantially more accurate than raw-IMU output branches, while DiT and latent diffusion expose different accuracy-plausibility tradeoffs. These results support pseudo-pose latent diffusion as an interpretable first-generation pipeline and identify heading stability, long-horizon rollout, and physics consistency as open problems.

Our contributions are fourfold. First, we introduce an IMU-only dog future-motion generation task with explicitly defined sensor input and motion output spaces. Second, we develop a reproducible DOGMA preprocessing pipeline that converts 20Hz raw IMU streams into structured pseudo-pose windows for visualization, modeling, and evaluation. Third, we quantify the representation gap between sensor-space and motion-space generation by comparing raw-IMU and pseudo-pose VAEs under the same evaluation protocol. Finally, we benchmark six conditional diffusion variants with shared prediction and plausibility metrics, providing empirical evidence for pseudo-pose-based generation as a practical first step toward scalable dog motion synthesis from wearable sensors.

2. Related Work

Animal motion capture and generation provide the broader context for this work, but existing methods do not address dog future-motion generation from body-worn IMUs. Dog pose datasets such as RGBD-Dog, 3DDogs, DigiDogs, BARC, and BITE study canine pose or shape from RGB, RGB-D, synthetic data, or motion-capture setups [6, 11–14]. These works show that dog pose estimation remains difficult even with visual supervision and controlled data. Recent animal motion generation methods, including OmniMotionGPT and AniMo, learn animal motion latent or token spaces and support text-driven generation [18, 19], but they do not condition on body-worn IMU streams. DOGMA-Gen instead targets an IMU-only setting that uses a derived pseudo-pose motion space for modeling and evaluation, without assuming motion-capture supervision or text labels.

Wearable IMU modeling is the closest technical area, but most prior work is human-centric. Sparse Inertial Poser and Deep Inertial Poser show that a small number of IMUs can recover full-body human motion only with strong temporal and body priors [5, 17]. PIP adds physical optimization to reduce artifacts from sparse inertial reconstruction [20], and recent calibration work highlights dynamic sensor-to-body misalignment in body-worn IMUs [21]. DiffusionPoser and PoseAugment introduce generative modeling into IMU-based human motion reconstruction or augmentation [8, 16]. These methods motivate our use of temporal priors and plausibility checks, but they assume human skeletons, human motion data, or physical models that are not directly available for dogs.

Latent generative models motivate the two-stage design

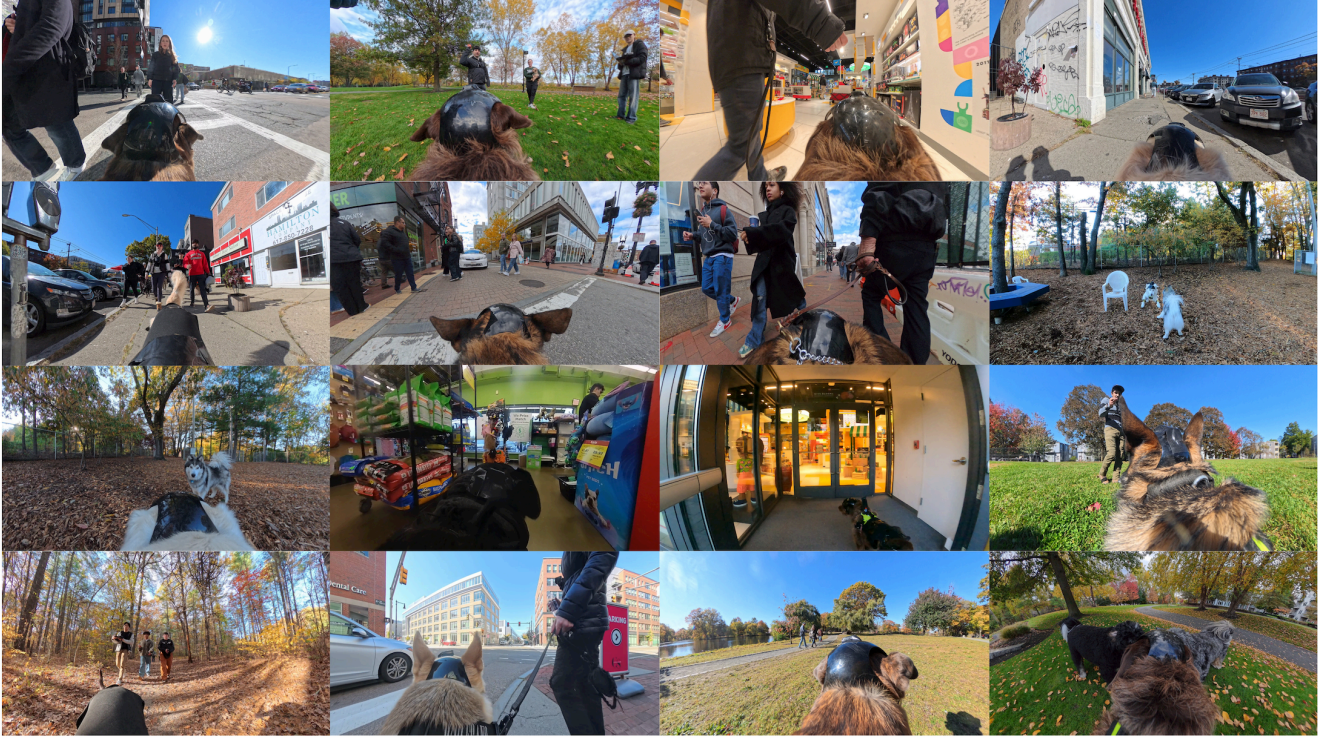


Figure 2. **DOGMA Data Qualitative Examples.** The source data consists of natural dog-human recordings with wearable cameras and body-mounted IMUs. We use the dog-side IMU stream as the conditioning signal and treat video and annotation as outside the current generation scope.

of our pipeline. VAEs provide probabilistic latent representations regularized by reconstruction and KL objectives [7]. Diffusion models learn iterative denoising processes for high-dimensional generation [4], and latent diffusion reduces cost by applying diffusion in a compressed representation [10]. In motion generation, MDM and MLD demonstrate the value of diffusion and latent diffusion for human motion synthesis [1, 15]. DiT replaces convolutional diffusion backbones with scalable transformer denoisers [9]. We adapt these ideas to dog IMU generation by using a temporal VAE over pseudo-pose windows and comparing convolutional latent diffusion with DiT-style data-space and latent-space baselines.

Because future motion is multimodal, evaluation should measure both conditional accuracy and distributional plausibility. MPJPE, heading error, ADE, and FDE compare a generated future with the held-out pseudo-pose future, but a single recorded continuation is not the only plausible continuation of a 6s IMU history. Motion generation work therefore often reports distributional metrics such as FID and diversity in learned motion-embedding spaces [2, 3, 18, 19]. We follow this practice with Motion-FID and Diversity@10, and add dog-specific plausibility metrics based on held-out real-motion jerk and range-of-motion bands. This combined protocol is necessary because

paired accuracy, diversity, and natural dog-like motion are distinct objectives.

3. Data and Representation

3.1. Dataset and Window Construction

We build on DOGMA, an ongoing naturalistic dog-human data collection effort in our lab with approximately 20 hours of synchronized video and IMU data from 19 dog-human pairs. Each dog wears a back-mounted 360° camera and 10 IMUs; the handler wears an egocentric camera and 8 IMUs. As shown in Fig. 2, the recordings cover sidewalks, parks, trails, indoor spaces, and other everyday environments, and include locomotion, turning, leash interaction, object interaction, sitting, lying, stair motion, and jumping. We use only the dog-side IMU stream and keep the 15 pairs with complete dog-side sensor coverage and enough valid windows. The remaining pairs are excluded because missing or unstable sensor streams would make pseudo-pose construction and participant-disjoint evaluation unreliable.

The original IMU recordings are sampled at 40 Hz and converted to a 20Hz IMU-only dataset. Downsampling reduces high-frequency jitter, storage, and training cost while preserving gait-scale dynamics for 6s history to 2s future prediction. Each pair of consecutive frames is

Table 1. **20Hz IMU Window Dataset and Participant Split.** The table reports participant splits, canonical 8 s segment counts, duration, and window coverage. The split is participant-disjoint and provides the 6s history plus 2s future used by the generation task.

Split	Pairs	Video segments	Total video (h:m:s)	Windows	Windowed (h:m:s)	Windowed / Video
Train	11	40	6:03:56	21,022	46:42:56	7.70×
Val	2	7	0:37:04	369	0:49:12	1.33×
Test	2	4	2:16:47	1,368	3:02:24	1.33×
Total	15	51	8:57:47	22,759	50:34:32	5.64×

merged into one 20Hz frame: quaternions are interpolated with midpoint SLERP, gyroscope and free-acceleration values are averaged, and interpolation flags are combined with a logical OR. Segments are filtered to retain recordings with all 10 dog sensors and sufficient valid duration. The canonical 8 s manifest contains 51 segments, with 40/7/4 train/validation/test segments and 21022/369/1368 windows, respectively. Table 1 summarizes the participant split, segment counts, duration, and window coverage. The split is participant-disjoint: 11 dog-human pairs for training, 2 for validation, and 2 for testing.

The generation task uses a past-future window within each 8 s clip. The condition is the first 6s, or 120 frames at 20Hz, and the target is the next 2s, or 40 frames. During VAE and latent-diffusion training, we represent each sample as the full 8 s pseudo-pose sequence so that the VAE latent encodes a temporally coherent context-plus-future clip. During evaluation and video export, only the final 40 frames are scored and rendered as the generated future.

3.2. Raw IMU Representation

The raw IMU input is a sensor-space sequence with 10 sensors and 13 features per sensor. For each sensor s and frame t , the feature vector is

$$\mathbf{x}_{t,s} = [\text{rot6d}(\mathbf{q}_{t,s}), \boldsymbol{\omega}_{t,s}, \mathbf{a}_{t,s}, m_{t,s}] \in \mathbb{R}^{13}, \quad (1)$$

where $\mathbf{q}_{t,s}$ is the orientation quaternion converted to a 6D rotation representation, $\boldsymbol{\omega}_{t,s} \in \mathbb{R}^3$ is gyroscope angular velocity, $\mathbf{a}_{t,s} \in \mathbb{R}^3$ is free acceleration, and $m_{t,s}$ is a binary interpolation flag. Flattening all sensors gives $\mathbf{x}_t \in \mathbb{R}^{130}$. A 6s condition window is therefore $\mathbf{X}_{1:120} \in \mathbb{R}^{120 \times 130}$.

This representation stays close to the measured signal. It preserves the wearable sensor information available at inference time and does not rely on video, text annotations, or external motion-capture labels. Because raw IMU is not directly interpretable as dog pose, the pipeline uses pseudo-pose as a derived motion space for visualization, VAE targets, and evaluation. Table 2 summarizes the raw-IMU, pseudo-pose, future-output, and VAE latent dimensions.

3.3. Pseudo-Pose Representation

Pseudo-pose is a structured motion representation derived from dog IMUs. Each frame contains 10 stern-centered rel-

Table 2. **Input, Output, and Latent Representation Dimensions.** Raw IMU is the condition space. Pseudo-pose is the motion space used for visualization, VAE targets, and reported metrics.

Representation	Per Frame	Window Shape
Raw IMU	$10 \times 13 = 130$	120×130
Pseudo-pose	$10 \times 3 + 6 + 3 = 39$	160×39
Future pseudo-pose	39	40×39
VAE latent	64	20×64

ative 3D joint positions, a 6D root heading representation, and a 3D root translation:

$$\mathbf{y}_t = [\mathbf{p}_{t,1}, \dots, \mathbf{p}_{t,10}, \mathbf{h}_t, \mathbf{r}_t] \in \mathbb{R}^{39}, \quad (2)$$

where $\mathbf{p}_{t,j} \in \mathbb{R}^3$ is a stern-centered relative joint position, $\mathbf{h}_t \in \mathbb{R}^6$ is the 6D representation of the stern/root heading rotation, and $\mathbf{r}_t \in \mathbb{R}^3$ is the root translation (x, y, z) at frame t . The joint positions are centered at the stern and should be interpreted as a derived pseudo-pose representation rather than measured 3D dog joints; the root translation provides the trajectory component used by ADE and FDE. This representation makes failures visible: heading flips, limb distortions, trajectory drift, and unstable gaits are easier to diagnose in a rendered skeleton than in raw accelerometer signals. It also provides a compact target for VAE and diffusion training and defines a common evaluation space across pose-space and raw-IMU ablations.

4. Method

4.1. System Overview

DOGMA-Gen is an IMU-only conditional motion-generation pipeline. The input is a raw IMU history window

$$\mathbf{C} = \mathbf{X}_{1:120} \in \mathbb{R}^{120 \times 130}, \quad (3)$$

corresponding to 6s at 20Hz. The desired output is a plausible future pseudo-pose sequence

$$\hat{\mathbf{Y}}_{121:160} \in \mathbb{R}^{40 \times 39}, \quad (4)$$

corresponding to the next 2s. Raw IMU is the conditioning signal because it is the sensor stream available at inference time. Pseudo-pose is the output space because it is

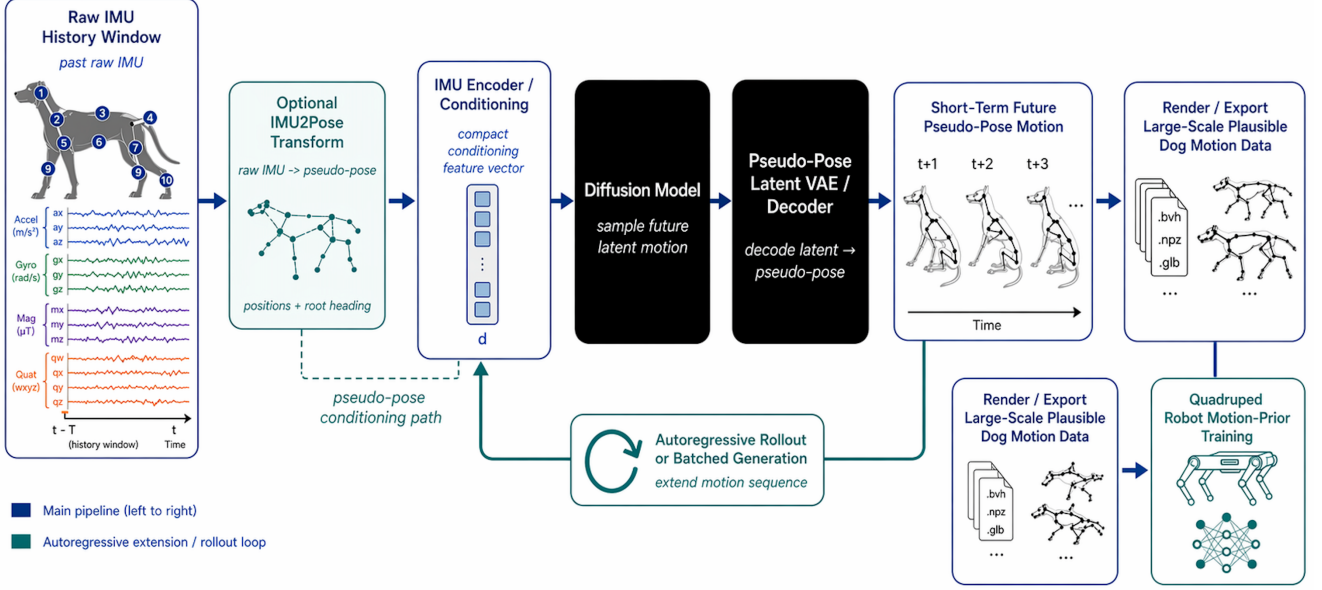


Figure 3. **Raw-IMU-Conditioned Future Motion Pipeline.** The model observes a 6s raw IMU history window. The IMU2Pose transform defines the pseudo-pose branch, and the learned generator returns a 2s future motion clip for visualization, evaluation, or export.

compact, visually inspectable, and better aligned with dog-motion plausibility than raw accelerometer or gyroscope reconstruction. Fig. 3 summarizes the raw-IMU-to-future-motion dataflow.

The pipeline has five stages. First, DOGMA IMU streams are synchronized, downsampled, filtered, and converted into 20Hz windows. Second, a deterministic IMU2Pose transform derives stern-centered pseudo-pose windows from the same raw IMU measurements. Third, temporal VAEs learn compact sequence latents for pseudo-pose and, for ablation, raw IMU. Fourth, conditional diffusion models sample future motion conditioned on the past raw IMU history. Fifth, generated short clips are rendered, evaluated, concatenated through autoregressive rollout, and exported as motion data.

4.2. IMU-to-Pose Transformation

The IMU2Pose module is a kinematic visualization transform rather than a learned pose estimator or force-based model. It converts raw dog IMU orientations into a structured, body-frame pseudo-pose that can be rendered and used as a target representation. Fig. 4 shows the orientation-normalization and rendering dataflow.

Let $q_{t,s}$ be the measured orientation quaternion for sensor s at frame t , and let r denote the stern/root sensor. Each sensor orientation is first expressed relative to the stern:

$$q_{t,s}^{\text{rel}} = q_{t,r}^{-1} \otimes q_{t,s}, \quad (5)$$

where $\otimes$ denotes quaternion multiplication. A segment-level neutral orientation is estimated from the sequence me-

dian or calibration window, and the neutralized relative rotation is applied to a rest vector $\mathbf{b}_s$:

$$\mathbf{p}_{t,s} = R(q_{t,s}^{\text{rel}} \otimes (q_s^{\text{neutral}})^{-1}) \mathbf{b}_s. \quad (6)$$

The resulting positions are relative to the stern. Root translation is stored as a 3D trajectory $\mathbf{r}_t = (x_t, y_t, z_t)$ for each frame. Root heading is extracted from the stern quaternion by projecting the root forward direction onto the ground plane:

$$\mathbf{f}_t = \frac{\Pi_{xy}(R(q_{t,r})\mathbf{e}_x)}{\|\Pi_{xy}(R(q_{t,r})\mathbf{e}_x)\|_2}, \quad (7)$$

and stored as a 6D rotation representation. This design keeps local joint motion, root heading, and short-horizon root translation together without treating the result as motion-capture-grade dog pose.

4.3. Temporal VAE

The temporal VAE compresses an 8 s pseudo-pose clip $\mathbf{Y}_{1:160} \in \mathbb{R}^{160 \times 39}$ into a latent sequence. The encoder is a temporal convolutional network with three stride-2 down-sampling stages, producing a latent trajectory with 20 time steps and 64 channels:

$$q_\phi(\mathbf{z} | \mathbf{Y}) = \mathcal{N}(\mu_\phi(\mathbf{Y}), \text{diag}(\sigma_\phi^2(\mathbf{Y}))), \quad \mathbf{z} \in \mathbb{R}^{20 \times 64}. \quad (8)$$

Sampling uses the reparameterization trick,

$$\mathbf{z} = \mu_\phi(\mathbf{Y}) + \sigma_\phi(\mathbf{Y}) \odot \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (9)$$

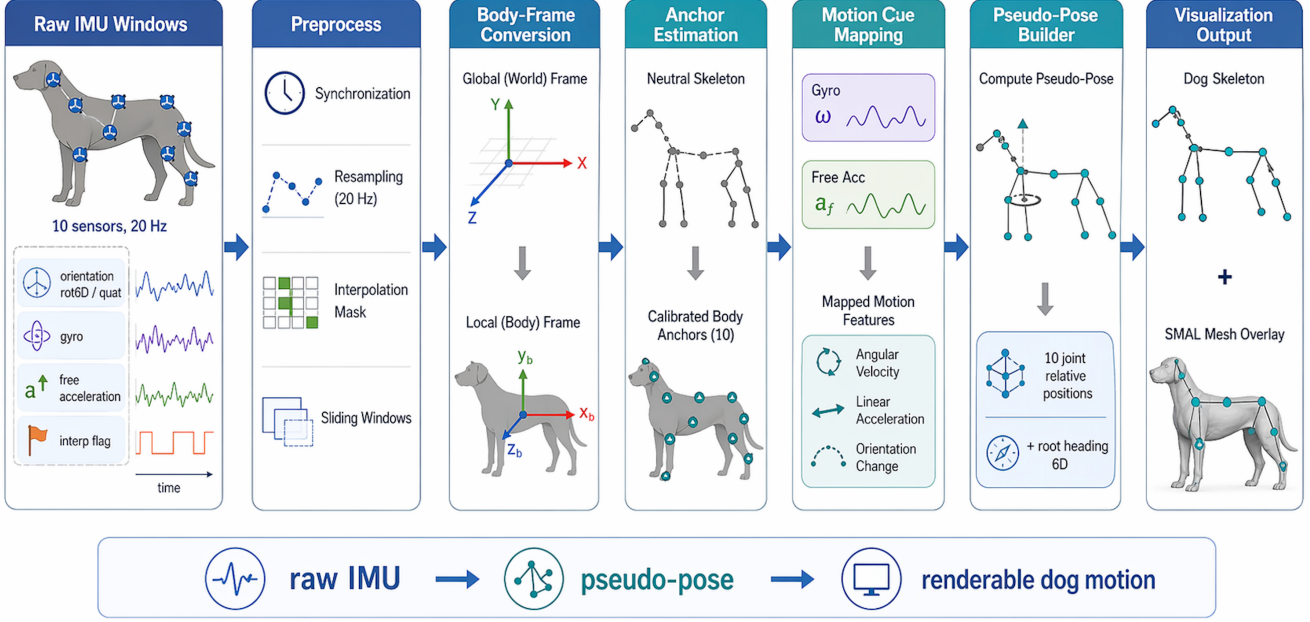


Figure 4. **IMU2Pose Pseudo-Pose Construction.** Raw IMU orientations are normalized in the stern/root frame, converted into relative segment rotations, combined with neutral rest vectors, and rendered as pseudo-pose motion. This module provides a common interpretation layer for visualization and pseudo-pose-space modeling.

and the decoder reconstructs $\hat{\mathbf{Y}} = D_{\theta}(\mathbf{z})$ in the original 160-frame pseudo-pose space.

The VAE objective combines relative-position reconstruction, root-heading reconstruction, root-translation reconstruction, optional smoothness terms, and KL regularization:

$$\begin{aligned} \mathcal{L}_{\text{VAE}} &= \lambda_p \|\hat{\mathbf{P}} - \mathbf{P}\|_2^2 + \lambda_h \|\hat{\mathbf{H}} - \mathbf{H}\|_2^2 \\ &\quad + \lambda_t \|\hat{\mathbf{T}} - \mathbf{T}\|_2^2 + \lambda_s \mathcal{L}_{\text{smooth}} + \beta \mathcal{L}_{\text{KL}}, \quad (10) \\ \mathcal{L}_{\text{KL}} &= \text{KL}(q_{\phi}(\mathbf{z} | \mathbf{Y}) \| \mathcal{N}(\mathbf{0}, \mathbf{I})). \end{aligned}$$

Here $\mathbf{P}$ denotes the relative joint-position sequence, $\mathbf{H}$ denotes the root-heading sequence, and $\mathbf{T}$ denotes the 3D root-translation sequence. The translation term is a mean-squared reconstruction loss over the per-frame root translation (x, y, z) , matching the translation component used later for ADE and FDE. The smoothness term is optional and is applied to temporal differences of the reconstructed motion when enabled. The raw-IMU VAE ablation has the same temporal role but reconstructs raw IMU windows instead of pseudo-pose windows. This ablation tests whether an IMU-output latent space can be projected back to plausible dog motion.

4.4. Conditional Diffusion Models

The conditional diffusion models generate future motion from the raw IMU condition $\mathbf{C}$. For pseudo-pose latent diffusion, the target is the VAE latent sequence $\mathbf{z}$ for the

8 s pseudo-pose clip. The forward diffusion process adds Gaussian noise:

$$\mathbf{z}_t = \sqrt{\bar{\alpha}_t} \mathbf{z}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon, \quad \epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I}), \quad (11)$$

and the denoiser learns

$$\mathcal{L}_{\text{diff}} = \mathbb{E}_{\mathbf{z}_0, t, \epsilon} \left[\|\epsilon - \epsilon_{\theta}(\mathbf{z}_t, t, \mathbf{C})\|_2^2 \right]. \quad (12)$$

Although the reported generation target is the final 2s future, the diffusion target is the latent of the full 8 s context-plus-future clip. This choice follows the frozen VAE interface: the VAE is trained on 160-frame pseudo-pose windows and produces a 20-step latent trajectory, so denoising the full latent lets a single decoder reconstruct a temporally coherent clip across the observed-to-future boundary. The past 6s portion therefore participates in the noise-prediction objective in Eq. 12, which is averaged over all latent time steps and channels. It is not reported as generated output: paired prediction metrics, videos, and exported samples use only frames 121–160. This does not leak future information at inference time, because the denoiser condition is only the past raw IMU window $\mathbf{C} = \mathbf{X}_{1:120}$; future pseudo-pose appears only as the supervised training target $\mathbf{z}_0$. At test time, reverse diffusion starts from Gaussian latent noise conditioned on $\mathbf{C}$, the frozen VAE decoder maps the sample to $\hat{\mathbf{Y}}_{1:160}$, and the final 40 frames are retained as the generated future.

We normalize both condition and target spaces using training-split statistics. The raw-IMU condition is z-scored

per feature dimension, and VAE latents are z-scored per latent channel across training windows and latent time steps. In the convolutional latent-diffusion denoiser, the noisy latent $\mathbf{z}_t \in \mathbb{R}^{20 \times 64}$ is projected with a 1×1 temporal convolution. The raw-IMU condition $\mathbf{C} \in \mathbb{R}^{120 \times 130}$ is projected to the same hidden width, encoded by temporal residual convolution blocks with stride-2 downsampling, and adaptively pooled to the 20 latent time steps. A sinusoidal diffusion-step embedding is passed through an MLP and added to the hidden sequence. For the reported convolutional models, condition fusion uses additive fusion,

$$\mathbf{h} = \mathbf{h}_z + \mathbf{h}_c + \mathbf{h}_t, \quad (13)$$

followed by residual temporal denoising blocks and a final 1×1 projection that predicts ϵ . We use a linear DDPM beta schedule from $\beta_1 = 10^{-4}$ to $\beta_T = 0.02$. The canonical convolutional latent-diffusion baseline uses hidden width 128, 4 residual denoising blocks, and 50 diffusion steps; the stronger Pose Latent-Diff branch in the six-model comparison uses hidden width 256, 12 residual denoising blocks, and 16 diffusion steps.

As summarized in Table 4, we compare six generator families under the same 6s history / 2s future protocol. Pose latent diffusion uses a convolutional temporal denoiser in the pseudo-pose VAE latent space. Raw-IMU latent diffusion uses the corresponding raw-IMU latent space and then projects outputs into pseudo-pose for comparison. Four DiT-style variants replace the denoiser with transformer diffusion models and test both data-space and latent-space targets for pseudo-pose and raw-IMU outputs. In all cases, raw IMU history is the condition; the generated target space and denoising backbone define the ablation.

The DiT-style baselines use the same past-IMU condition and full-window target convention, but inject the condition through transformer tokens rather than convolutional additive fusion. The normalized 120-frame raw-IMU history is embedded as condition tokens, the noisy data-space or latent-space target is embedded as target tokens, and the two token sequences are concatenated with fixed 1D sine-cosine position embeddings and learned token-type embeddings. The timestep embedding modulates each transformer block through adaptive layer normalization, and the final layer predicts noise only for the target tokens. The DiT branches in Table 4 use hidden width 128, depth 4, 8 attention heads, a linear beta schedule from 10^{-4} to 0.02, and 100 diffusion steps.

4.5. Motion Encoder for Embedding

Distributional metrics require a feature space in which generated and real future clips can be compared. We train a motion encoder adapted from Guo et al. [2] on pseudo-pose clips and use its frozen embeddings for Motion-FID and Diversity@10. As shown in Fig. 5, each input clip

is first mapped by a linear layer, then encoded by a bidirectional GRU. The final forward and backward states are concatenated and passed through an MLP to produce the motion embedding. A projection head with linear, normalization, GELU, and L2-normalization layers is used only during contrastive training and is discarded for evaluation.

For a pseudo-pose clip, two augmented views are encoded and projected to normalized vectors $\mathbf{z}_i$ and $\mathbf{z}_i^+$. With cosine similarity $\text{sim}(\cdot, \cdot)$ and temperature τ , the contrastive objective is

$$\mathcal{L}_{\text{enc}} = \frac{1}{B} \sum_{i=1}^B -\log \frac{\exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_i^+)/\tau)}{\sum_{k \neq i} \exp(\text{sim}(\mathbf{z}_i, \mathbf{z}_k)/\tau)}. \quad (14)$$

This training encourages embeddings of two views from the same motion clip to remain close while separating embeddings from different clips. After training, we use the encoder embedding before the projection head as $\psi(\cdot)$.

4.6. Autoregressive Rollout and Export

The core model is trained for short-horizon generation because 2s futures are a tractable target under IMU drift and pseudo-pose uncertainty. Longer clips can be obtained by autoregressive rollout: after generating $\hat{\mathbf{Y}}_{121:160}$, the system advances the window and reconditions on the next available or predicted history. In the current implementation, rollout is used for export and qualitative inspection rather than as a fully validated long-horizon model. Short clips are evaluated quantitatively, while long rollouts are used to inspect temporal stability, drift, and gait plausibility before future physics-aware evaluation.

5. Experiments

5.1. Experimental Setup

All experiments use the IMU-only 20Hz dataset described in Sec. 3. The participant split is fixed and disjoint across train, validation, and test. The VAE experiments train on 21022 windows and validate on 369 windows, and reconstruction is reported on all 1368 test windows. The six-model generation comparison uses 1364 scored test windows after evaluator-specific validity filtering. Diversity@10 is computed on the first 200 test windows with 10 samples per condition for tractability. Table 3 summarizes the shared training and evaluation settings.

The pseudo-pose VAE uses hidden size 128, latent size 64, KL weight $\beta = 10^{-3}$, and a transpose-convolution decoder. The canonical latent diffusion run uses a frozen 100-epoch VAE, past length 120, future length 40, and a 6s history to 2s future evaluation window. The six-model comparison additionally includes a stronger pose latent-diffusion run with hidden size 256 and 12 temporal blocks, plus four DiT branches trained with 100 diffusion steps. Models

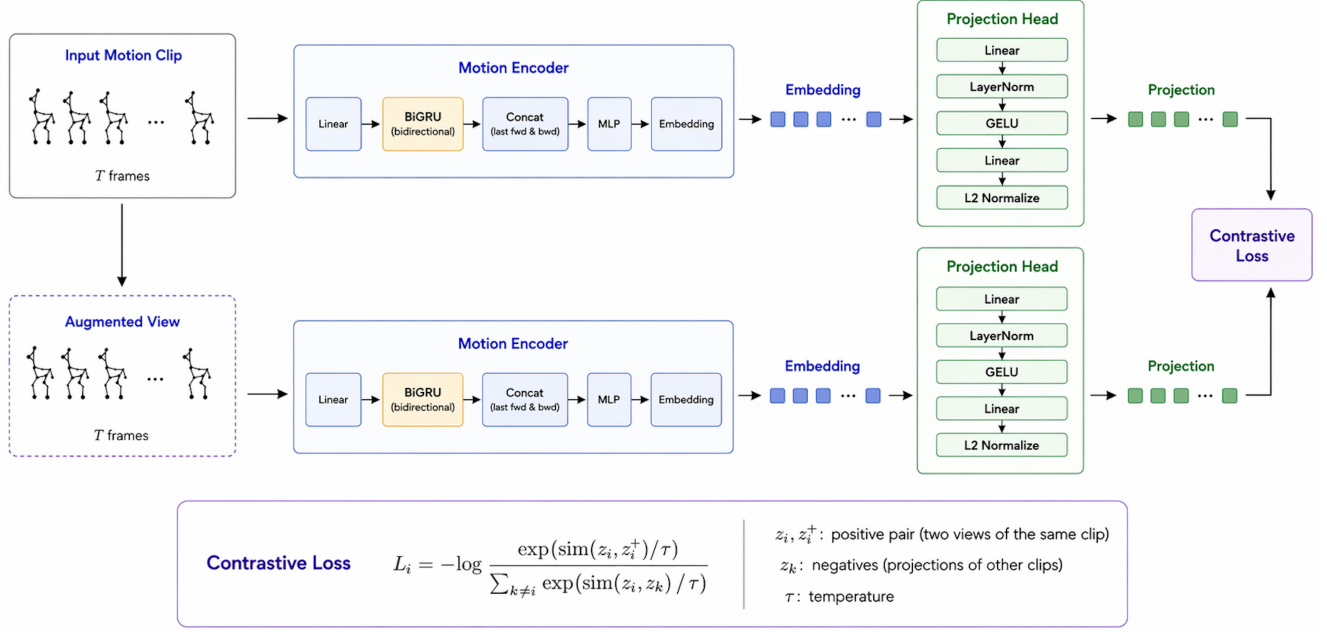


Figure 5. **Motion Encoder.** Motion-FID and Diversity@10 are computed after embedding future clips with a frozen motion encoder adapted from Guo et al. [2], the metrics compare generated and real clips in a learned motion feature space rather than raw coordinate space alone.

Table 3. **Training Protocol and Evaluation Setup.** All models use the same canonical 6s history to 2s future task. Training time reports single-GPU wall-clock or GPU-hours where logged or estimable from run artifacts.

Model	Input / condition	Target	Frames	Latent	Main optimizer setting	Training time
Pseudo-pose VAE	pseudo-pose clip	pseudo-pose clip	160	20×64	100 epochs, batch 16, lr 5×10^{-4}	1.20 GPU-h
Raw-IMU VAE	raw IMU clip	raw IMU clip	160	20×64	100 epochs, batch 16, lr 5×10^{-4}	1.28 GPU-h
Pose latent diffusion	raw IMU history	pseudo-pose latent	120→160	20×64	100 epochs, batch 16, 16–50 steps	0.56–0.82 GPU-h
DiT variants	raw IMU history	data or latent target	120→40/160	varies	300 epochs, batch 64, 100 steps	2.00 GPU-h

are trained on three NVIDIA RTX 3090 GPUs. A single pseudo-pose VAE run takes about 1.20 GPU-hours, the raw-IMU VAE takes about 1.28 GPU-hours, and the canonical pose latent-diffusion baseline takes about 0.56 GPU-hours. The full-train latent-diffusion capacity searches used 18-hour wall-clock budgets over three GPUs; selected candidates take roughly 0.73–0.82 GPU-hours each. DiT training takes about 2.00 GPU-hours.

5.2. Model Comparison and Ablations

We ablate two design choices: whether the output should be pseudo-pose or raw IMU, and whether diffusion should operate in data space or VAE latent space. Table 4 lists the six generator variants.

The VAE ablation compares a pseudo-pose VAE with a raw-IMU VAE that reconstructs sensor features directly. The diffusion ablation repeats the pose-space versus raw-space comparison for conditional generation.

5.3. Evaluation Metrics

Let $\mathbf{P}, \hat{\mathbf{P}} \in \mathbb{R}^{T \times J \times 3}$ denote held-out pseudo-pose targets and generated relative joint positions for the future window. Mean per-joint position error (MPJPE) is

$$\text{MPJPE} = \frac{1}{TJ} \sum_{t=1}^T \sum_{j=1}^J \|\hat{\mathbf{p}}_{t,j} - \mathbf{p}_{t,j}\|_2. \quad (15)$$

Heading error compares root forward directions computed from the stern/root quaternion:

$$\text{HE} = \frac{180}{\pi T} \sum_{t=1}^T \arccos\left(\text{clip}\left(\hat{\mathbf{f}}_t^\top \mathbf{f}_t, -1, 1\right)\right), \quad (16)$$

where $\mathbf{f}_t$ is defined in Eq. 7. We additionally evaluate root-trajectory prediction with Average Displacement Error (ADE) and Final Displacement Error (FDE). Let $\mathbf{r}_t, \hat{\mathbf{r}}_t \in \mathbb{R}^3$ be the held-out pseudo-pose target and generated root translations at future frame t . ADE measures the mean Eu-

Table 4. **Generator Ablations Across Target and Denoising Spaces.** All methods condition on the same past raw IMU history. The target space and denoiser differ.

Model name in report	Target space	Generative space	Purpose
Pose Latent-Diff	pseudo-pose	VAE latent	Main interpretable pipeline (UNet-based diffusion model)
Raw-IMU Latent-Diff	raw IMU	raw-IMU VAE latent	Tests whether IMU output space is viable
Pose DiT Data	pseudo-pose	data space	Tests transformer diffusion without VAE compression
Pose DiT Latent	pseudo-pose	VAE latent	Tests transformer denoising in pose latent space
Raw-IMU DiT Data	raw IMU	data space	Raw sensor-space DiT baseline
Raw-IMU DiT Latent	raw IMU	raw-IMU VAE latent	Raw sensor-space latent DiT baseline

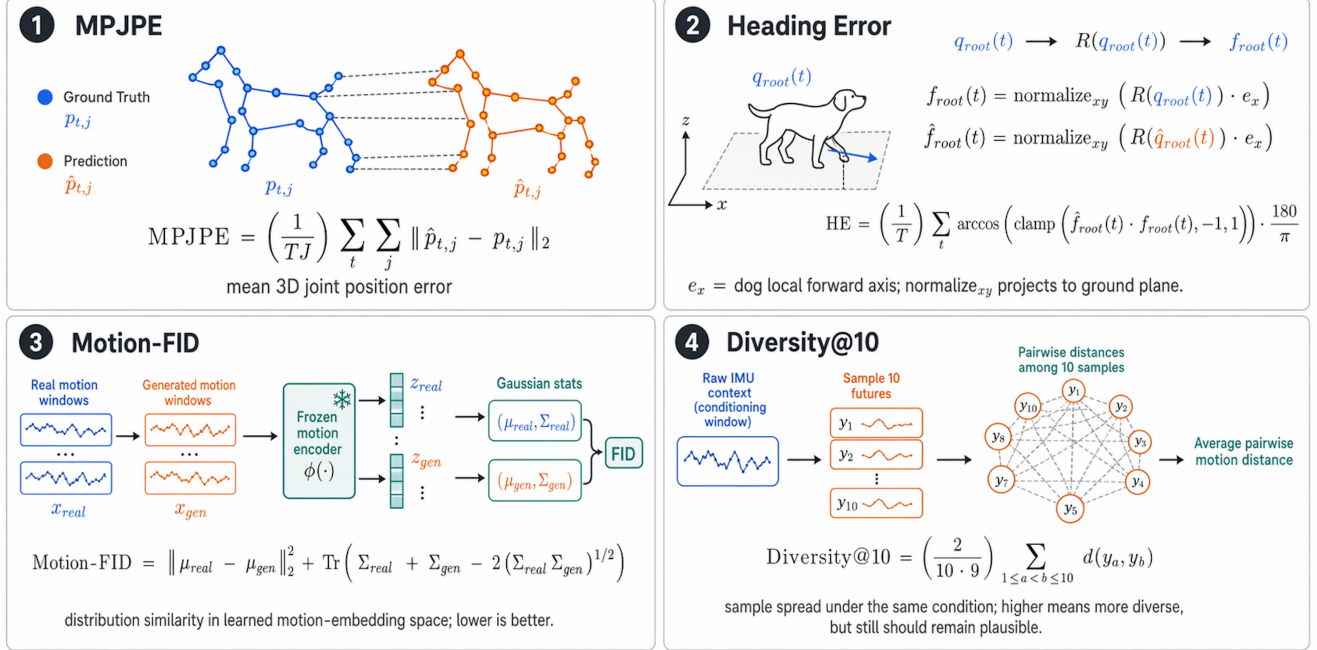


Figure 6. **Evaluation Metrics.** Evaluation combines paired future error, trajectory error, distributional metrics, and plausibility metrics. MPJPE, heading error, ADE, and FDE measure alignment with the held-out future; Motion-FID and Diversity@10 measure realism and sample spread in a learned motion-embedding space.

clidean root-translation error over the full future window:

$$\text{ADE} = \frac{1}{T} \sum_{t=1}^T \| \hat{\mathbf{r}}_t - \mathbf{r}_t \|_2, \quad (17)$$

FDE measures the Euclidean root-translation error at the final predicted frame:

$$\text{FDE} = \| \hat{\mathbf{r}}_T - \mathbf{r}_T \|_2. \quad (18)$$

Together, ADE captures average trajectory accuracy while FDE captures endpoint drift. These paired metrics compare generated motion against the recorded future, so lower is better.

For generation quality evaluation, generated and real future clips are embedded with the frozen motion encoder in-

roduced in Sec. 4.5. Motion-FID follows the Fr chet distance between Gaussian fits to real and generated motion-encoder features:

$$\text{FID} = \| \mu_r - \mu_g \|_2^2 + \text{Tr} \left(\Sigma_r + \Sigma_g - 2(\Sigma_r \Sigma_g)^{1/2} \right). \quad (19)$$

Diversity@10 measures the average pairwise distance among $K = 10$ generated samples for the same condition:

$$\text{Div@10} = \frac{2}{K(K-1)} \sum_{1 \leq a < b \leq K} \mathbf{d}_{ab}, \quad K = 10, \quad (20)$$

$$\mathbf{d}_{ab} = \| \psi(\hat{\mathbf{Y}}^{(a)}) - \psi(\hat{\mathbf{Y}}^{(b)}) \|_2.$$

where $\psi(\cdot)$ is the frozen motion encoder.

The plausibility metrics reported for the six-model comparison are Motion-FID, Diversity@10, jerk-band viola-

Table 5. **Conditional Generation Results.** MPJPE is converted to millimeters. ADE and FDE are Euclidean root-translation errors in pseudo-pose root units: ADE averages over the 40-frame future window, and FDE uses the final future frame. Prediction and plausibility metrics use 1364 windows; Diversity@10 uses 200 conditions with 10 samples each. Lower is better except Diversity@10, whose real-motion reference is 19.55.

Model	MPJPE (mm) ↓	Heading (°) ↓	Motion-FID ↓	Div@10	Jerk-V ↓	ROM-V ↓	ADE (root) ↓	FDE (root) ↓
Pose Latent-Diff	92.27	38.15	24.77	19.33	0.001	0.004	1.996	1.985
Raw-IMU Latent-Diff	188.42	75.83	15.65	19.42	0.609	0.600	5.042	6.681
Pose DiT Data	86.79	38.08	14.67	19.18	0.238	0.013	1.699	1.689
Pose DiT Latent	84.80	38.67	15.34	19.28	0.518	0.006	1.913	1.906
Raw-IMU DiT Data	198.73	54.62	20.87	18.15	0.999	0.460	5.195	7.181
Raw-IMU DiT Latent	166.19	57.06	16.08	18.75	0.225	0.401	8.343	11.335

tion, and range-of-motion (ROM) band violation. For a scalar plausibility statistic $g_m(\mathbf{Y})$ with held-out real-motion quantile band $[l_m, u_m]$, the violation rate is

$$\text{Viol}_m = \frac{1}{N} \sum_{i=1}^N \mathbb{I} \left[g_m(\hat{\mathbf{Y}}_i) < l_m \text{ or } g_m(\hat{\mathbf{Y}}_i) > u_m \right]. \quad (21)$$

In Eq. 21, m indexes the plausibility statistic, N is the number of generated clips, and $\mathbb{I}[\cdot]$ contributes one when a generated clip falls outside the real-motion band for that statistic. The bounds l_m and u_m are computed once from held-out real motions, so the metric measures whether generated samples remain within the empirical range of realistic dog motion rather than whether they match a specific future frame. A violation rate of 0 means every generated clip lies inside the real-motion band for metric m ; larger values indicate more frequent high-frequency jitter or implausible limb configurations. These band metrics are not a substitute for physics simulation, but they expose artifacts that may not dominate MPJPE.

6. Results and Analysis

6.1. Generative Model Results

Table 6 shows that pseudo-pose is a substantially stronger reconstruction target than raw IMU. The pseudo-pose VAE reconstructs held-out motion with 7.55 mm MPJPE and 1.69° heading error over all test windows. The raw-IMU VAE, after projecting its reconstructed sensor sequence into pseudo-pose, yields 129.92 mm MPJPE and 39.16° heading error. Thus, the raw branch loses motion structure needed for visualization and downstream plausibility, even when it reconstructs a sequence in the measured sensor space.

6.2. VAE Reconstruction Results

The high-quality validation clip from the Cynthia-WolfGang pair shows the same pattern qualitatively. The pseudo-pose VAE overlay follows the held-out pseudo-pose target closely, while the raw-IMU VAE projection exhibits visible limb and heading mismatch. This supports the main

Table 6. **VAE Reconstruction on Held-Out Test Windows.** Both models are evaluated on the same 1368 test windows and reported in pseudo-pose space.

Model	MPJPE (mm) ↓	Heading (°) ↓
Pseudo-pose VAE	7.55	1.69
Raw-IMU VAE	129.92	39.16

representation choice: raw IMU is a useful conditioning signal, but pseudo-pose is a more reliable target for motion synthesis.

Table 5 shows three main trends. First, pseudo-pose targets outperform raw-IMU targets on paired future accuracy. The best pose-space MPJPE is 84.80 mm, compared with 166.19 mm for the best raw-IMU branch. Heading error follows the same pattern: pose-space branches cluster around 38°, whereas raw-IMU branches range from 54.62° to 75.83°. Root-trajectory errors also favor pose-space outputs, with the best ADE/FDE achieved by Pose DiT Data. Although the absolute errors remain large, the gap is consistent and representation-driven.

Second, no single model dominates all metrics. Pose DiT Latent gives the lowest MPJPE, Pose DiT Data gives the best Motion-FID, heading error, ADE, and FDE, and Pose Latent-Diff gives the lowest jerk and ROM violation. The current system therefore does not have a single best operating point. Latent compression, transformer denoising, and direct data-space denoising trade off local accuracy, root-trajectory alignment, distributional realism, and high-frequency smoothness.

Third, raw-IMU output branches can appear competitive on some distributional statistics while still failing as motion generators. Raw-IMU Latent-Diff has Diversity@10 closest to the real-motion reference, but its paired error and plausibility-band violations are poor. Sensor-space diversity alone is therefore insufficient: generated IMU must also project to a structured and plausible dog-motion representation.

Table 7. **Condition and Future Window Length Ablation.** Pose DiT Data is evaluated with different raw-IMU condition lengths and future prediction horizons.

Model	Condition	Future	MPJPE (mm) ↓	Heading (°) ↓	Motion-FID ↓	Div@10	ADE (root) ↓	FDE (root) ↓
Pose DiT Data	6 seconds	2 seconds	86.79	38.08	14.67	19.18	1.699	1.689
	6 seconds	4 seconds	83.47	38.10	19.97	19.21	1.697	1.774
	4 seconds	2 seconds	82.84	31.94	11.91	19.43	1.661	1.713
	4 seconds	4 seconds	84.44	34.07	17.87	19.20	1.632	1.715
	2 seconds	2 seconds	82.11	26.99	12.25	19.44	1.491	1.658
	2 seconds	4 seconds	83.73	29.60	13.20	19.41	1.571	1.636

6.3. Analysis

Table 7 shows that shorter conditioning windows are not necessarily weaker for this task. The 2s-condition models give the best MPJPE, heading error, diversity, and root-trajectory errors, while the 4s-to-2s setting gives the best Motion-FID. This result is plausible because the most recent IMU frames carry the strongest information about gait phase, current heading, and short-term root velocity.

The future horizon has a smaller but still visible effect. Extending prediction from 2s to 4s does not consistently degrade MPJPE, but it tends to worsen Motion-FID and final displacement under longer conditioning. This suggests that local pose accuracy and distributional realism respond differently to horizon length: the model can keep average joint error low over a longer window, yet the generated sequence may drift away from the real motion distribution or accumulate endpoint error. Overall, the length ablation supports using recent IMU context for short-horizon generation rather than assuming that more history is always beneficial.

The representation comparison in Tables 6 and 5 points to the same practical conclusion: target representation matters more than simply increasing denoiser capacity. A raw-IMU latent can encode sensor trajectories, but projection from reconstructed sensors to pseudo-pose amplifies calibration errors, orientation drift, and local sensor noise. Pseudo-pose latents directly encode the quantities used for visualization and evaluation, which explains why the pose-space branches are more reliable across paired motion and root-trajectory metrics.

The data-space versus latent-space comparison is more nuanced. In Table 5, Pose DiT Data achieves the best Motion-FID, suggesting that direct denoising better matches the real future distribution in the learned motion-encoder space. Pose DiT Latent achieves the best MPJPE, suggesting that the VAE latent regularizes paired prediction. The convolutional Pose Latent-Diff model has the strongest jerk and ROM plausibility despite worse FID, suggesting that architecture and training objective can bias outputs toward smoother motion. Fig. 7 further shows that deeper and wider latent-diffusion variants can improve some validation objectives but do not remove the need for multi-

metric model selection.

6.4. Discussion of Failure Modes

The main failure modes are heading drift, root-trajectory drift, unstable limb timing, and implausible gait transitions. Heading and translation are especially difficult because small root-orientation or root-position errors can accumulate visually; a generated sample may have plausible limb motion but face or move in the wrong direction. Raw-IMU target models additionally suffer from projection failures: small orientation or acceleration reconstruction errors can become large relative-position and trajectory errors after the IMU2Pose transform.

Longer rollout remains open. The current 2s future clips are short enough to evaluate against a held-out future, but autoregressive concatenation can compound heading and phase errors. The system should therefore be viewed as a short-term generation pipeline rather than a stable long-horizon dog simulator. Its current value is to identify which representation choices are viable for future scaling.

6.5. Qualitative Results

Fig. 8 shows the comparison between ground truth and predicted root translation in the horizontal plane. It shows 2s future root translation given past 6s as conditions. The predictions can follow the broad direction of travel while still drifting in curvature or endpoint location. This helps explain why ADE and FDE are useful alongside MPJPE and heading error. A motion can have plausible local limb movement but still be visually wrong if the root path bends or terminates incorrectly.

Qualitative videos provide an additional diagnostic for each model family. VAE reconstruction videos overlay reconstruction and target pseudo-pose motion to reveal whether the latent space preserves limb timing, root heading, and gait shape. Diffusion videos render generated 2s futures without target overlays to test whether samples are plausible on their own. The final six-model comparison renders the same segment with all six generators in a 2-by-3 grid, making representation-dependent failures easier to compare. These videos are available in the [Google Drive video folder](#).

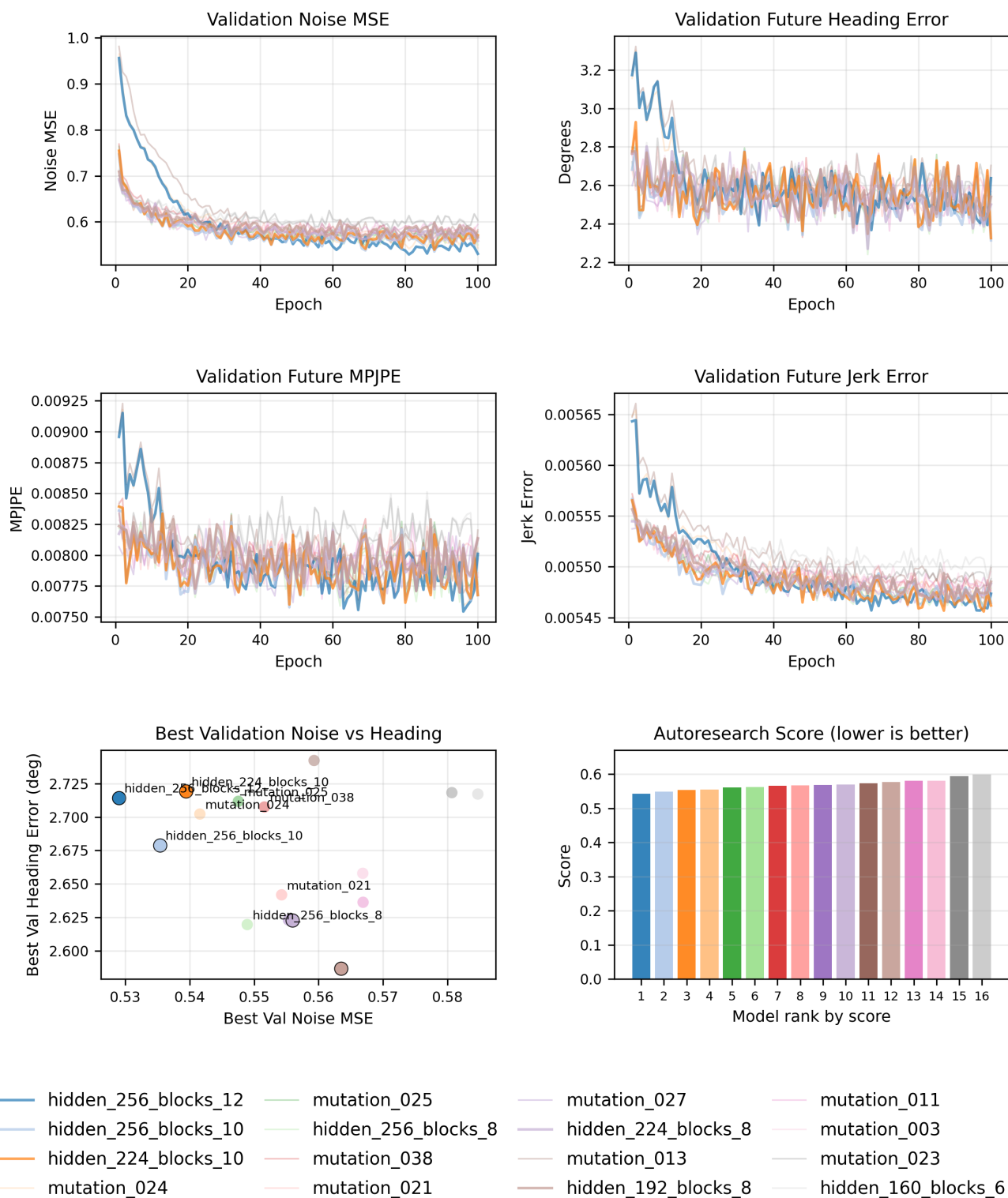


Figure 7. **Latent-Diffusion Capacity Ablation.** Hybrid depth/width variants improve some validation objectives but do not remove the need for multi-metric model selection.

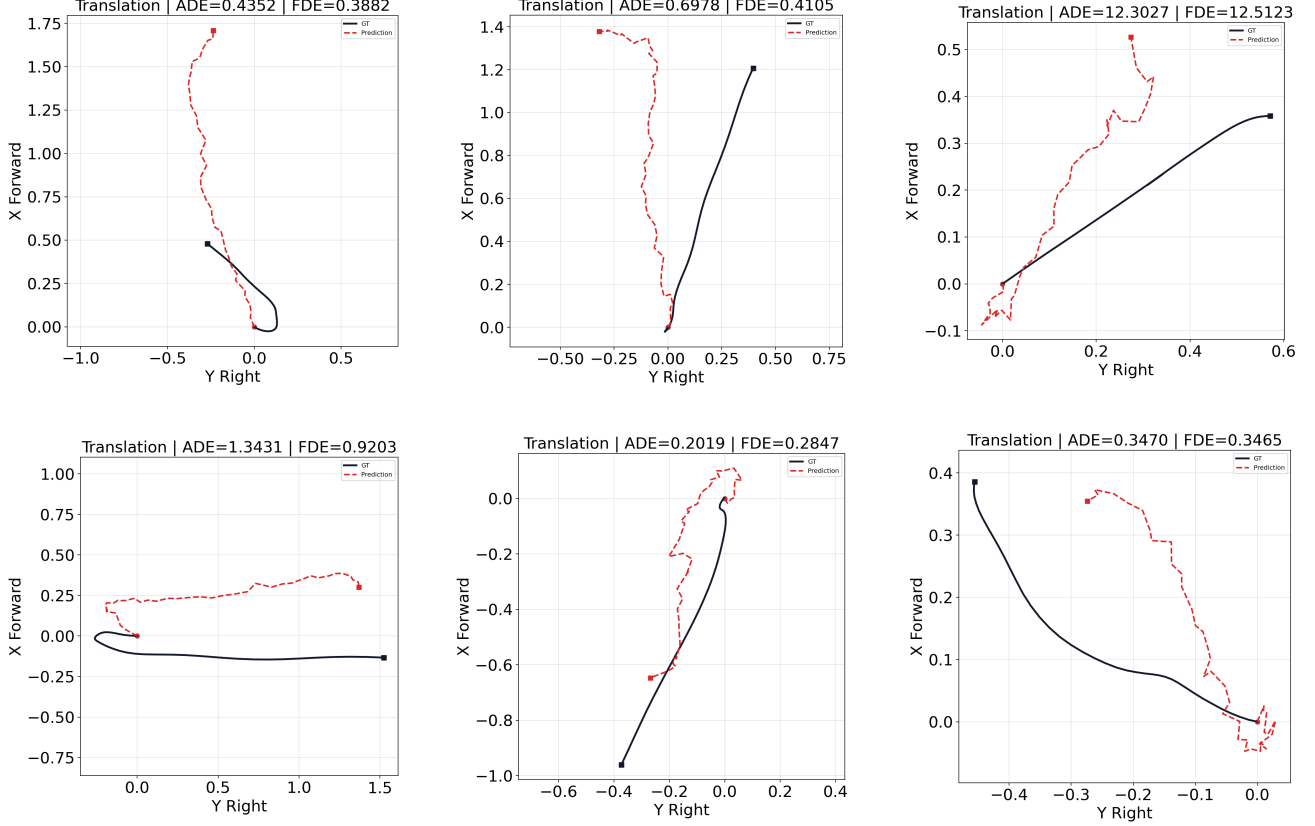


Figure 8. **Qualitative Results of Root-Translation.** We show root translation examples in the horizontal (x, y) plane with ADE and FDE.

7. Limitations and Future Work

7.1. Current Limitations

This work should be interpreted as an early IMU-conditioned dog motion generation benchmark rather than a complete dog motion simulator. Its main limitations come from data coverage, pseudo-pose fidelity, horizon length, and evaluation scope.

First, the dataset is still small relative to the diversity of dog motion. DOGMA is large for naturalistic dog wearable sensing, but 19 dog-human pairs cannot cover the full range of breeds, body sizes, gait styles, environments, harness attachments, and sensor drift patterns. The participant-disjoint split tests whether models generalize beyond the training identities, but it is not a substitute for broad cross-breed and cross-environment validation.

Second, pseudo-pose is an approximate derived representation. It is computed from IMU orientations through a stern-centered kinematic transform, not measured by optical motion capture. This makes it useful as a structured space for modeling, rendering, and evaluation, but it should not be interpreted as measured 3D dog pose. The transform depends on sensor placement, neutralization, rest vec-

tors, root-heading extraction, and root-translation estimation; loose attachment or unusual body configurations can therefore create systematic pseudo-pose errors before the generative model is trained.

Third, the current generators are short-horizon models. Quantitative results are reported for 2s futures, while longer rollouts are used only for qualitative inspection. Autoregressive stitching can accumulate heading, phase, and root-trajectory errors, so the current system should not be treated as a stable long-horizon dog simulator. The evaluation protocol also remains incomplete for physical validity: it measures relative pose, heading, root translation, distributional realism, diversity, and empirical jerk and range-of-motion bands, but not contact, force, balance, foot sliding, or robot-transfer performance.

7.2. Future Directions

The first priority is motion-representation improvement. A learned or hybrid IMU2Pose module could incorporate dynamic calibration, body-part decomposition, temporal filtering, and dog-specific kinematic constraints. The pseudo-pose representation could also be extended beyond stern-centered relative joints by modeling root translation, head-

ing, contacts, and gait phase more explicitly. Improving this representation would reduce the gap between raw wearable signals and generated motion, and would make downstream models less sensitive to sensor placement noise.

The second priority is model improvement for longer and more stable generation. Future generators should be trained for temporal consistency across overlapping windows rather than only stitched at inference time. Promising directions include recurrent conditioning, diffusion over longer latent windows, phase-aware motion priors, trajectory-aware losses for heading and root translation, and consistency losses between adjacent generated clips. These changes target the dominant visual failures in the current system: heading drift, endpoint drift, and gait-phase discontinuities.

The third priority is physical and robotics-facing evaluation. The current plausibility metrics are empirical diagnostics, not physical validation. Future work should add contact estimation, foot-skating checks, body-stability constraints, and simulation-based retargeting to quadruped robots. A robot-facing benchmark would make it possible to test whether generated dog motion improves policy learning, reward shaping, or motion-prior training, rather than only improving pseudo-pose-space metrics.

7.3. Applications

Large-scale Dog Motion Data Generation. The long-term goal is an IMU-only pipeline that turns wearable recordings into plausible dog motion clips. Once pseudo-pose construction and generation become more reliable, the same pipeline could expand DOGMA-like recordings into a larger synthetic dog motion corpus. Unlike manual animation or laboratory motion capture, this route could scale from in-the-wild sensing while preserving natural behavior statistics.

Quadruped Robot Motion-prior Training. Generated dog motion should not be copied directly into a robot controller because dog morphology, contact mechanics, and actuation differ from robotic quadrupeds. Its more realistic role is as a motion prior: generated clips may provide gait-phase distributions, turning patterns, posture transitions, and short-horizon regularizers. These priors could support imitation pretraining, reward shaping, latent motion priors, or data augmentation before policy learning in physics simulation. The goal is not to make a robot reproduce every dog pose, but to test whether robust animal-inspired motion structure can improve robot learning.

8. Conclusion

We proposed DOGMA-Gen, a reproducible pipeline for IMU-only dog future-motion generation from wearable sensors. Given 6s of raw dog-worn IMU history, the task is to generate a plausible 2s future clip in a derived pseudo-pose space. The pipeline connects 20Hz DOGMA preprocess-

ing, IMU2Pose pseudo-pose construction, temporal VAE modeling, conditional latent diffusion and DiT-style generation, and shared prediction and plausibility evaluation.

We found that representation choice matters more than simply increasing generator capacity. Pseudo-pose latents preserve the derived motion representation far better than raw-IMU latents projected back to pseudo-pose space, and pose-space diffusion branches are consistently stronger paired predictors than raw-IMU target branches. At the same time, the six-model comparison shows that MPJPE, root-trajectory error, Motion-FID, smoothness, and diversity define different operating points rather than a single dominant model.

DOGMA-Gen should therefore be viewed as a first benchmark and design study, not as a complete dog motion simulator. The next steps are to improve IMU-to-pseudo-pose modeling, train rollout-aware generators for longer horizons, and evaluate physical validity through contact-aware and robotics-facing tests. Progress on these directions could turn in-the-wild wearable recordings into a scalable source of animal motion priors for dog motion synthesis and quadruped robot learning.

References

- [1] Xin Chen, Biao Jiang, Wen Liu, Zilong Huang, Bin Fu, Tao Chen, Jingyi Yu, and Gang Yu. Executing your commands via motion diffusion in latent space. *arXiv preprint arXiv:2212.04048*, 2023. 3
- [2] Chuan Guo, Shihao Zou, Xinxin Zuo, Sen Wang, Wei Ji, Xingyu Li, and Li Cheng. Generating diverse and natural 3d human motions from text. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 5152–5161, 2022. 3, 7, 8
- [3] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local nash equilibrium. In *Advances in Neural Information Processing Systems*, 2017. 3
- [4] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. In *Advances in Neural Information Processing Systems*, pages 6840–6851, 2020. 3
- [5] Yinghao Huang, Manuel Kaufmann, Emre Aksan, Michael J. Black, Otmar Hilliges, and Gerard Pons-Moll. Deep inertial poser: Learning to reconstruct human pose from sparse inertial measurements in real time. *ACM Transactions on Graphics*, 37(6):185:1–185:15, 2018. 1, 2
- [6] Sinead Kearney, Wenbin Li, Martin Parsons, Kwang In Kim, and Darren Cosker. RGBD-Dog: Predicting canine pose from rgbd sensors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2020. 2
- [7] Diederik P. Kingma and Max Welling. Auto-encoding variational bayes. *arXiv preprint arXiv:1312.6114*, 2013. 3
- [8] Zhuojun Li, Chun Yu, Chen Liang, and Yuanchun Shi. Poseaugment: Generative human pose data augmentation with physical plausibility for IMU-based motion capture. In *European Conference on Computer Vision*, 2024. 2
- [9] William Peebles and Saining Xie. Scalable diffusion models with transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pages 4195–4205, 2023. 3
- [10] Robin Rombach, Andreas Blattmann, Dominik Lorenz, Patrick Esser, and Björn Ommer. High-resolution image synthesis with latent diffusion models. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 10684–10695, 2022. 3
- [11] Nadine Rueegg, Silvia Zuffi, Konrad Schindler, and Michael J. Black. BARC: Learning to regress 3d dog shape from images by exploiting breed information. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2022. 2
- [12] Nadine Rueegg, Shashank Tripathi, Konrad Schindler, Michael J. Black, and Silvia Zuffi. BITE: Beyond priors for improved three-d dog pose estimation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023.
- [13] Moira Shooter, Charles Malleson, and Adrian Hilton. Benchmarking monocular 3d dog pose estimation using in-the-wild motion capture data. *arXiv preprint arXiv:2406.14412*, 2024.
- [14] Moira Shooter, Charles Malleson, and Adrian Hilton. DigiDogs: Single-view 3d pose estimation of dogs using synthetic training data. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision*, 2024. 2
- [15] Guy Tevet, Sigal Raab, Brian Gordon, Yonatan Shafir, Daniel Cohen-Or, and Amit H. Bermano. Human motion diffusion model. *arXiv preprint arXiv:2209.14916*, 2022. 3
- [16] Tom Van Wouwe, Seunghwan Lee, Antoine Falisse, Scott Delp, and C. Karen Liu. Diffusionposer: Real-time human motion reconstruction from arbitrary sparse sensors using autoregressive diffusion. *arXiv preprint arXiv:2308.16682*, 2023. 2
- [17] Timo von Marcard, Bodo Rosenhahn, Michael J. Black, and Gerard Pons-Moll. Sparse inertial poser: Automatic 3d human pose estimation from sparse imus. *Computer Graphics Forum*, 36(2):349–360, 2017. 1, 2
- [18] Xuan Wang, Kai Ruan, Xing Zhang, and Gaoang Wang. AniMo: Species-aware model for text-driven animal motion generation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2025. 2, 3
- [19] Zhangsihao Yang, Mingyuan Zhou, Mengyi Shan, Bingbing Wen, Ziwei Xuan, Mitch Hill, Junjie Bai, Guo-Jun Qi, and Yalin Wang. OmniMotionGPT: Animal motion generation with limited data. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024. 2, 3
- [20] Xinyu Yi, Yuxiao Zhou, Marc Habermann, Soshi Shimada, Vladislav Golyanik, Christian Theobalt, and Feng Xu. Physical inertial poser (PIP): Physics-aware real-time human motion tracking from sparse inertial sensors. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pages 13167–13178, 2022. 1, 2
- [21] C. Zuo, J. Huang, X. Jiang, Y. Yao, X. Shi, R. Cao, X. Yi, F. Xu, S. Guo, and Y. Qin. Transformer IMU calibrator: Dynamic on-body IMU calibration for inertial motion capture. In *ACM SIGGRAPH Conference Papers*, pages 1–14, 2025. 1, 2